EE3400

Electronic Measurements and Instrumentation



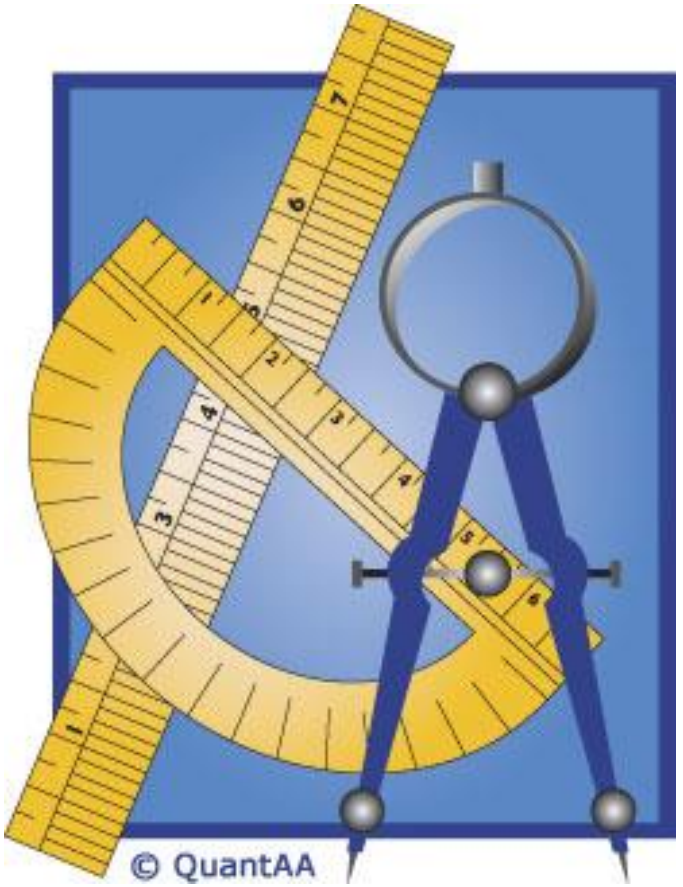
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Overview of the Course

- What is instrumentation?
- General overview of basic concepts in measurement
- Different measurement methods:
  - Resistive Methods; Capacitive Methods; Inductive Methods; Ultrasonic Methods; Digital Methods
- Design of sensors for measurement of various measurands
  - Length; displacement; velocity; force; level; pressure; flow; temperature

Measurement

- It is the process of sampling and quantifying a physical variable e.g. temperature
- It is achieved by comparing an unknown variable to a standard unit and finding the ratio between the two



Instrumentation



- Is the physical hardware used to make a measurement e.g. a thermometer
- There are a wide variety which utilize many different physical properties

Transducers, Actuators and Transponders

- Transducer/Sensor:
  - converts a physical variable into an electrical signal
- Actuator:
  - turns an electrical signal into a physical action
- Transponder:
  - turns physical variables into electrical signals and visa versa



Why is all of this Important?

- Measurement has been vital ever since humans started to make or design anything
- It forms the backbone of all experimental procedure and engineering design
- Current and future backbone of 4th industrial revolution – Autonomous robots and cars, Smart energy or cities,etc

Role of Instrumentation Engineer



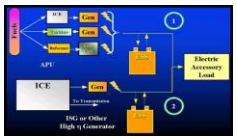
Primary Measurand

SENSORS

Interface

Chemical

Electrical



PROCESS

CONTROLLER



Mechanical

ACTUATORS

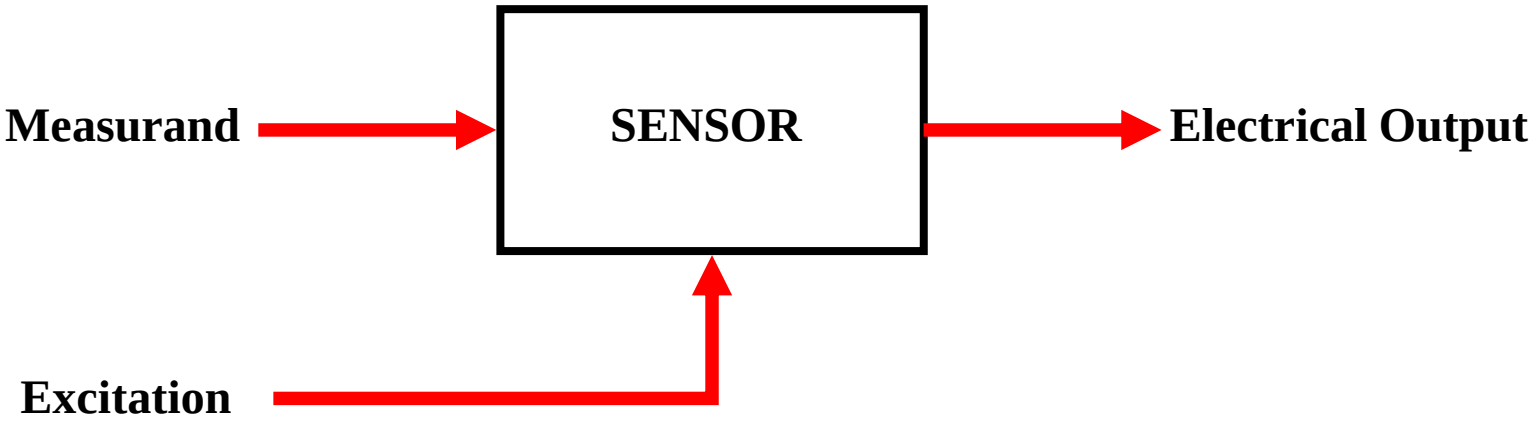
Interface



Interface

Sensors

- The function of a sensor is to:
  - convert a signal from one physical form to another
  - provide us with an electrical output which can be used to quantify the variable of interest
- A measurand is the physical property, condition or quantity which a sensor converts into an electrical signal



What does a Sensor Measure?

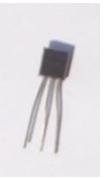
- Sensors often do not measure the physical variable of interest directly, but rather another property of the system which represents the measurand
- For example if we wish to measure temperature we can measure:
  - The change in resistance of a piece of wire (Platinum resistance thermometer, PRT)



  - The thermoelectric potential generated by a junction between two dissimilar conductors (Thermocouple)



  - The change in the band gap of a doped semiconductor (a semiconductor temperature sensor)



- In this example, temperature is known as the *primary measurand*. The other variables that we measure are known as secondary and tertiary measurands.

Primary Measurands

- Examples of some primary measurands are:

- Position
- Velocity
- Acceleration
- Force
- Sound Amplitude
- Sound Frequency
- Electrical Potential
- Electrical Current Flow
- Electrical Charge
- Electrical Frequency
- Magnetic Flux Density
- Magnetic Flux Intensity
- Strain
- Light (EM Radiation) Amplitude
- Light (EM Radiation) Frequency
- Temperature
- Heat Radiation
- Heat Flux
- Flow Rate
- Viscosity
- Density
- Altitude
- Altitude Rate
- Specific Gravity
- Torque
- Stress

Strain Gauge Example

- A strain gauge's *primary measurand* is the amount of strain placed on a material
- Changes in strain produce changes in the resistance of the gauge, so therefore electrical resistance is our *secondary measurand*
- In order to measure resistance of the gauge we can either put:
  - A constant voltage across it and measure the changes in current
  - A constant current through it and measure the changes in the voltage across it
- Therefore our *tertiary measurand* is either current or voltage

Measurement Criteria
Standard Unit

- A standard unit is decided upon by a community and is defined in terms of an actual object or condition
- It is the unit by which all devices design for measuring that quantity are calibrated

- The International System of units (SI) is used to define most units in Engineering (Distance measured in meters(m))
- It is the modern form of the metric system
- Devised on the base number 10

SI Units Table

Units in Common Use in the English and SI System

Quantity	English			SI		
	Name	Symbol	Units	Name	Symbol	Units
Frequency	Hertz			Hertz	Hz	s^{-1}
Energy	Foot-pound	ft·lb	$lb \cdot ft^2/s^2$	Joule	J	$kg \cdot m^2/s^2$
Force	Pound	lb	$lb \cdot ft/s^2$	Newton	N	$kg \cdot m/s^2$
Resistance	Ohm			Ohm	Ω	$kg \cdot m^2$ per $(s^3 \cdot A^2)$
Electric Potential	Volt			Volt	V	$A \cdot \Omega$
Pressure	Pound per in^2	psi	lb/in^2	Pascal	Pa	N/m^2
Charge	Coulomb			Coulomb	C	A·s
Inductance	Henry			Henry	H	$kg \cdot m^2$ per $(s^2 \cdot A^2)$
Capacitance	Farad			Farad	F	$s^4 \cdot A^2$ per $(kg \cdot m^2)$
Magnetic flux				Weber	Wb	V·s
Power	Horsepower	hp	$lb \cdot ft^2/s^3$	Watt	W	J/s

Measurement Criteria
Fundamental & Derived Quantities

- There are three fundamental quantities:
  - Length
  - Mass
  - Time
- All other quantities can be derived from these
- E.g.:

$$F = ma$$
$$= \left(kg \cdot m / s^2\right) = Newton \left[N\right]$$

How to Choose a Sensor?

- When choosing a sensor for a particular application we need to consider the following:
  - Is the measurement feasible?
  - What are the accuracy, range and bandwidth constraints?- Measurement criteria
  - What are the cost implications?
  - What kind of interfaces do we require?



Is the Measurement Practical or Feasible?

- Most measurements are feasible if you are willing to spend large amounts of money, time and effort to get them
- Therefore the first step in a project is to determine whether the measurement is feasible with regards to time and budget
- Does a commercial sensor exist which will fulfil the design criteria? Don’t reinvent the wheel
- If not, or if it is too expensive, make sure it is possible to make a cheaper version

Measurement Criteria
Accuracy

- Is the degree to which a measured value conforms to the actual value
- To calculate accuracy, we need to know the measurement accuracy error or error rate
- The measurement error rate is a percentage of the difference of the observed value to the actual value

$$\%Accuracy = 100 - \%Error_rate$$

$$\%Error_rate = \frac{|Measured\ Value - Actual\ Value|}{Actual\ Value} \times 100$$

- For example if a temperature sensor gave an output of 47ºC when the actual temperature was 50ºC, the error rate of the sensor would be:

$$\%Errorrate = \frac{|47 - 50|}{50} \times 100 = \underline{6\%}$$

$$\%Accuracy = 100 - 6 = 94\%$$

Measurement Criteria
Range, Span and Reproducibility

- The range of a sensor is the lower and upper bounds of value that a sensor can measure.
  - e.g. a thermometer whose scale goes from -10°C to 50°C has range from -10°C to 50°C.
- The span of a sensor is the distance between the lower and upper bounds of the sensor.
  - e.g. 60°C for the thermometer above
- Reproducibility is the ability of a sensor to give the same measurement for the same value of measurand, repeatedly.

Measurement Criteria
Precision

- Precision of the sensor refers to the limits within which a signal can be read and may be somewhat subjective.
  - E.g. the analogue instrument is graduated in 0.2psi, hence needle position can be approximated to within 0.02psi, therefore precision is 0.02psi.



- In other terminology, it's a degree of reproducibility

Measurement Criteria
Resolution

- This is usually only given for digital sensors
- It describes the smallest variation in the measurand that the sensor can measure

$$Resolution = \frac{Span}{2^{N}}$$

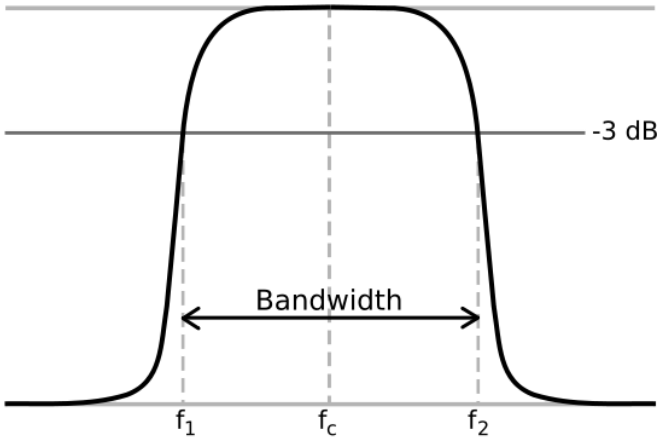
- For example, if we consider an LM35 temperature sensor (Sensitivity 10mV/ºC) which is digitised by a 16-bit ADC over a 5V span, we find it has a resolution of

$$\text{Resolution} = \frac{\text{Span}}{2^{N}}$$

$$= \frac{5V}{2^{16}} = \frac{5}{65536} = 76.3\mu V$$

Measurement Criteria
Bandwidth

- The bandwidth of a sensor is the range of frequencies within which it can follow variations in the measurand
- Described as the range between the -3dB points in the instruments frequency response.
- For example the human ear has a bandwidth of approximately 20Hz – 20kHz



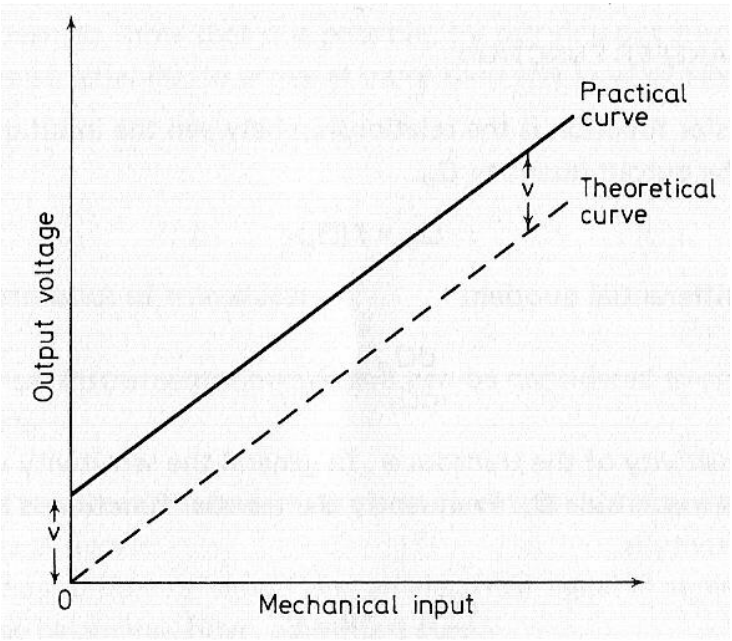
Measurement Criteria
Sensitivity & Responsiveness

- Sensitivity is the steady-state transfer function of the sensor i.e. ratio of how much output for how much input
- If we again look at the LM35 temperature sensor, it has a sensitivity of 10mV/ºC which gives a 10mV change in output for a change of 1ºC in temperature on the input
- Responsiveness is the percentage change in a signal required to obtain any change in the output

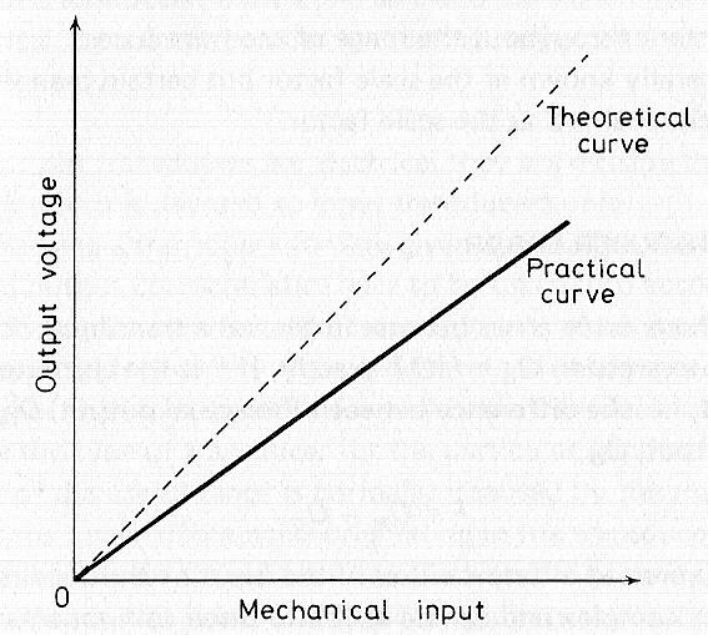
Measurement Criteria

Error

Sensor Zero Error



Sensor Sensitivity Error



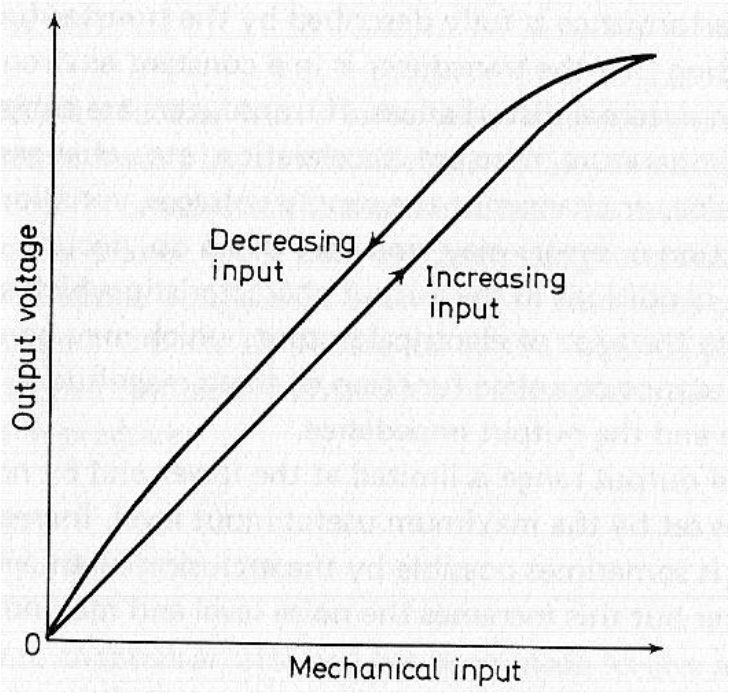
Mansfield, P.H. (1973) Electrical Transducers for Industrial Measurement

- Error is the difference between the expected value and the actual value of the measurand which can either be static or dynamic

Measurement Criteria
Hysteresis

- A sensor's inability to measure the same value in the upward and downward directions

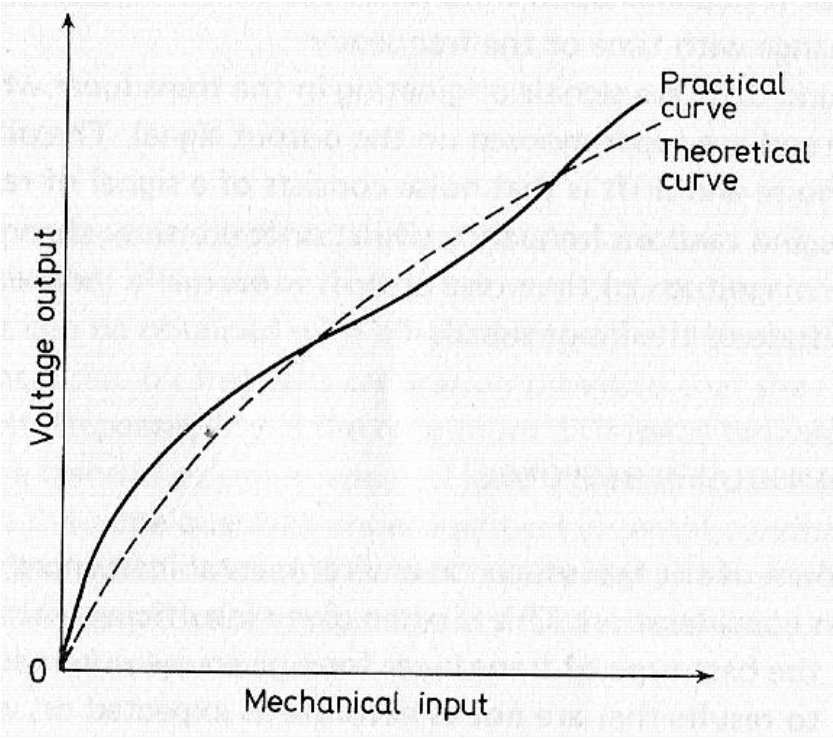
Sensor Hysteresis



Mansfield, P.H. (1973) Electrical Transducers for Industrial Measurement

Measurement Criteria
Non-Conformity

Sensor Non-Conformity



Mansfield, P.H. (1973) Electrical Transducers for Industrial Measurement

- This is the sensor output deviation from the theoretical transfer function
- In the case of a linear system, a deviation from the curve is known as non-linearity

What are the Cost Implications?

- As most measurements are feasible, cost is one of the major deciding factors in sensor selection
- Off-the-shelf sensors usually reduce the amount of time taken in design
- Sensor design is a time consuming process and should only be considered where a commercial product is not suitable for the task



What kind of Interfaces do we Require?

- There are many different options on the market today

- Basically a choice between
  - An analogue output (0-10V perhaps) or a current range of (4-20mA)
  - Digital output in the form of either a serial or parallel data stream, or maybe a simple yes/no(binary) output